

David W Miller

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Software Engineer, Unreal Engine & Cinematic Content Tools

Software Engineer with 10+ years building production-scale tools, content pipelines, and real-time systems for AAA game development. Deep hands-on expertise in Unreal Engine 4/5 (Blueprints, C++, Python, Plugin Development) with a strong track record designing cinematics workflows, animation tools, and DCC integrations for cross-disciplinary art teams. Shipped tools across 15+ AAA titles including Fortnite, New World, and multiple cinematic trailers. Equally fluent in engineering rigor and artist empathy, able to gather requirements, define technical direction, mentor teammates, and deliver maintainable, scalable systems under production pressure.

TECHNICAL SKILLS

Pipeline Engineering & Development

Python (Pipeline Tools, PyQt/PySide, PIL, FFMpeg, REST/CRUD APIs), C++, Perforce Helix Core, Git/GitHub Actions, Shotgun, SyncSketch, Docker, Jenkins, MongoDB, AWS Deadline, Visual Studio/VSCode, PyCharm, Jira, Confluence

Game Engine & Real-Time Rendering

Unreal Engine 4.x–5.x (Blueprints, C++, Python, Plugin Development, Materials/Shaders, Unreal Insights, Morph Targets, Grooms, Control Rig, LODs), Physically Based Rendering (PBR) & Stylized Materials, NVIDIA Nsight Graphics, Real-time Optimization, Motion Capture (Vicon Shogun, Optitrack Cameras)

3D Content Creation & Technical Art

Autodesk Maya (MEL, M.A.S.H, Bifrost. Rigging, Blendshapes, mGear, nCloth, XGen), SideFX Houdini (Environment Art & FX), ZBrush, Mudbox, Texturing (Adobe Substance 3D/Designer/Painter), Adobe Creative Suite (Photoshop, Illustrator, After Effects, Premiere)

EXPERIENCE

Pipeline Engineer | Halon Entertainment | Mar 2021 - Mar 2026

Custom materials and shader development, building pipeline tools, and production automation systems supporting 20+ Unreal Engine projects spanning AAA game cinematics, commercials and film. Defined technical direction for distributed art teams, architected scalable toolsets, and provided hands-on engineering support across character, environment, and sequencer-based production.

Fortnite -- Epic Games (Chapters 3, 4, 6, 7 Seasonal Cinematics)

- Authored Python toolset to standardize Level Sequencer organization across all seasonal cinematic productions — conforming play ranges, synchronizing subsequences, and locking tracks to enforce pipeline consistency across large distributed teams.
- Built utility scripts to detect and resolve broken physics assets on characters within complex Sequencer hierarchies, reducing artist debugging time significantly across multi-season productions.
- Developed Blueprints supporting lighting artists and character workflows; created post-process materials and custom shaders deployed across seasonal content releases.
- Provided hands-on Perforce Helix Core and UnrealGameSync (UGS) support, resolving workspace conflicts and sync failures to keep artists unblocked during live production.

New World — Amazon Games (Announce Trailer, Teaser Trailer & In-Game Intro Cinematic)

- Built Blueprint systems for dynamic lighting and character control pipelines used across multiple cinematic deliverables including the launch announce trailer, teaser, and the shipped in-game intro cinematic.
- Developed shaders for complex materials including glass, volumetric fog, and stylized water surfaces for real-time cinematic rendering in Unreal Engine.
- Created Cascade particle effects integrated into cinematic sequences to support environment and atmosphere storytelling.

Borderlands 3, Tiny Tina's Wonderlands & Remnant 2

- Developed gestalt character assembly pipeline for cinematic productions — parsing game asset part lists and constructing complete cinematic-ready characters that could be directly handed off to animators and riggers.
- Built a turntable renderer system for Remnant 2's gestalt inventory system, allowing layout artists to visually scan creatures and environment assets to inform scene composition decisions.
- Delivered shader development, utility scripting, and artist support across all three productions, working directly within shipped game asset libraries.

ESPN SportsCenter HQ — LED Wall Virtual Environment

- Architected and integrated a real-time Unreal Engine environment for ESPN's SportsCenter HQ LED wall broadcast infrastructure, connecting level geometry and shaders to the production control room via a custom plugin.
- Implemented live control room integrations enabling operators to dynamically assign graphics to geometry (billboards, signage), trigger geometry animation, and swap between HDRI lighting environments and time-of-day states via sublevels — all live during broadcast.

Other Projects

- Cinematic trailer work on Vampire the Masquerade: Bloodlines 2, KSP2, contributing shader development, character pipeline support, and sequencer-based technical art.
- Designed and implemented in-house branch of open-source AYON Unreal Engine plugin, managing distributed asset workflows, metadata synchronization, and pipeline orchestration across multiple projects.
- Deployed and maintained AWS Deadline distributed render farm, automating job scheduling, resource allocation, and failure recovery at production scale.

Technical Artist | Halon Entertainment | Oct 2017 - Mar 2021

- Lead Technical Artist on previsualization for Call of the Wild and other films managing skinning, rigging, establishing art workflows and providing technical support for environment, animation and lighting artists.
- Provided technical art support for numerous virtual production shoots, troubleshooting real-time rendering issues, camera tracking integration, and LED wall calibration during live capture sessions.
- Optimized Unreal Engine environments for Intel and AMD graphics card demos, implementing LOD strategies, reducing material complexity and draw calls, and tuning lighting and post-processing to hit performance targets for retail showcase deployment.
- Collaborated with asset and character artists, animators, designers, and engineers to optimize real-time assets, integrate complex scenes, and build custom materials while maintaining high visual fidelity under production deadlines.

Engine Artist & Game Developer | MatterVR | Feb 2017 - Aug 2017

- Shipped mobile VR game Wonders of the World for GearVR using early Unreal Engine 4, handling full technical art pipeline from asset integration to performance optimization.
- Built custom VR rendering pipeline, profiling and optimizing materials, meshes, and lighting to maintain consistent 60fps targets across mobile hardware constraints.
- Developed real-time multiplayer minigames in BP & C++, implementing networked gameplay for historical board games with data-driven progression systems.
- Created modular material systems and reusable art components, establishing technical standards that improved asset creation efficiency across the project.

JSC Distance Learning Instructional Designer | NASA JSC, GHG Corporation July 2014 - Oct 2016

- Delivered +200 interactive online lessons to the Flight Operations Directorate software familiarization and classroom-based instruction, providing refresher briefings and training courses to flight controllers, instructors and astronauts.
- Developed customized training overviews for ISS laptop software by meeting with stakeholders and developers and hosting planning sessions.
- SharePoint developer & administrator, supporting instructors and division pages across the Training Standards & Integration Production Group.
- Prototyped a fully-functional ISS Environment in Unreal Engine 4 for Interactive Emergency Training.

JSC Distance Learning Training Standards Intern | NASA JSC, United Space Alliance | Sep 2011 - Jul 2014

- Built 3 ISS Emergency Response online training courses using Articulate Storyline for Fire, Depress or Toxic Atmosphere scenarios using training footage from station mockups.
- Camera Operator for Video Lab capture of Flight Controller and Astronaut training footage and briefings.

EDUCATION

M.A. Digital Media Studies / University of Houston | Dec 2016

Compositing, Illustration, 3D Modelling & Animation

B.A. Communications / University of Houston | May 2011

TV Production Studio, School Newspaper & Magazine